DRUMPANEL · DESIGN DECISION RECORD

DrumPanel R7 kept at 2.5 drums an hour

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DOCUMENT CONTROL

Document ID	DMP-DDR-004	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/decisions/0004-r7-kept-at-2-5-drums-an-hour.md @ de21d74		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Amish's round-2 decision 5A on R7 recorded

0004: R7 kept at 2.5 drums an hour; the TRL 4 timed trial decides

- **Date:** 2026-10-03
- **Status:** decided by Amish, 2026-10-03 (round 2): "i agree with all the 46 recommendations you provided. please proceed." For DrumPanel this is decision 5A, the open decision in DMP-DEC-001 v0.2.

Context

After the notching punch, the ring head and the dial adjuster (DMP-DDR-003) two people turn about 2.26 drums an hour on paper: 53.1 hands-on minutes a drum against the 48.0 that 2.5 drums an hour needs (DMP-CAL-001 v0.2, section 8). Most of the task times behind the estimate are first estimates, not measurements.

Options considered

Option	What it means
A (chosen)	Keep R7 at 2.5 drums an hour and let the TRL 4 timed half-day trial decide; no more hardware at TRL 3
B	Restate R7 to 2.25 drums an hour, what the design gives on paper
C	Keep 2.5 and add more speed-ups now (rail stops, a second deburring station), with no estimate of what they save

Decision

Option A. No design change. R7 stays at "at least 2.5 drums to flat sheet per hour for a two-person team" and is reported honestly as not met on paper (about 2.26). The half-day trial in the build plan (section 5, Throughput) records each task's time and the drums an hour, and decides R7. If the trial misses 2.5, options B and C come back to Amish with measured times.

Table 1. Effects.

Item	Result
R7	Not met on paper, 2.26 against 2.5 drums an hour; kept; the TRL 4 trial decides
Other requirements	None changed
Mass and geometry	Unchanged
Cost	Unchanged. Value-engineering target: USD 3,000. Estimated cost of the constructable design: USD 3,489 (USD 489 over the target). `budget_usd` unchanged

Safety

> **Safety:** An empty oil drum can still hold flammable vapour and explode when cut, ground or heated. No drum is cut until it has been purged and passed the vapour check (build plan, stops S4 and S5), and the throughput target never justifies skipping a stop. The kit has shear heads, a punch and a roller: guards stay on and hands stay clear. It is an open engineering reference, not certified equipment.